

RUDRA SAMADHIYA

Full-Stack Developer — React · React Native · Node / NestJS

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#) • [prepsharp.in](#)

EDUCATION

Indian Institute of Information Technology (IIIT), Bhopal

2025 – 2029

B.Tech, Computer Science & Engineering (Data Science) • CGPA: 8.6 / 10

The Sanskaar Valley School, Bhopal

2024

Class X & XII (CBSE) • XII: 91.25% • X: 92%

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, C++

Frontend: React, Next.js, React Native (Expo), Tailwind / NativeWind

Backend: Node.js, NestJS, REST APIs, WebSockets (Socket.IO), JWT Auth, Microservices & Distributed Systems

Databases & Cache: PostgreSQL (TypeORM), MongoDB, Firebase / Firestore, Redis

DevOps & Tools: Docker, Git & GitHub, Vercel, Razorpay, Expo EAS, VS Code

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, System Design

PROJECTS

Instant Kirana — Hyperlocal Quick-Commerce Platform

[GitHub → instant-kirana](#)

React Native (Expo), TypeScript, NestJS, PostgreSQL, MongoDB, Redis, Socket.IO, Docker

- Architected a **distributed** Blinkit-style quick-commerce platform for Tier-2/3 India: **3 React Native apps** (customer, kirana owner, delivery partner) + a web admin dashboard, backed by **7 NestJS microservices** (auth, store, product, order, earnings, media, notification).
- Built real-time order & delivery tracking over **Socket.IO** with infinite-retry reconnect and a 5s REST-polling fallback, so status never stalls on flaky 3G networks.
- Cut perceived latency with a multi-layer caching stack — Redis server cache, Vercel edge (stale-while-revalidate), a WebP image CDN, and client-side predictive prefetch — plus optimistic UI and skeleton loaders.
- Implemented weighted MongoDB full-text search (ReDoS-safe regex, field projection) and JWT auth with Razorpay payments.

PrepSharp.in — Computer-Based Test (CBT) Exam Platform

[Live → prepsharp.in](#)

Next.js, React, TypeScript, Firebase / Firestore, React Native (Expo), KaTeX, Chart.js

- Built and shipped a **live** web platform (prepsharp.in) + an Android app (in Google Play review) serving a **133,000+ question bank** and **241 full-length JEE Main & Advanced papers**, with timed tests, a scoring engine, answer review, and Chart.js analytics.
- Engineered an **automated PDF extraction pipeline** that parses chapter, difficulty, and answer-type metadata into a normalized Firestore question bank (exam → subject → chapter → question).
- Added Google Sign-In (expo-auth-session), shared KaTeX math rendering across web & mobile, and in-memory caching to cut repeat Firestore reads.

Point of Sale (POS) System — Full-Stack Java (Web + CLI)

GitHub → POSSystem

Java, JDBC, PostgreSQL, REST APIs (JDK HttpServer), Docker, GitHub Actions CI

- Built and deployed a multi-tenant POS & inventory platform in pure object-oriented Java (no frameworks) — a web dashboard + REST API on the JDK HttpServer plus an interactive CLI; live on Render at possystem-pvup.onrender.com.
- Engineered a transactional billing engine (server-side stock validation + rollback to prevent overselling) and a pluggable persistence layer swapping CSV ↔ PostgreSQL via JDBC behind one Repository interface, with GitHub Actions CI testing both backends.

ACHIEVEMENTS & ACTIVITIES

- Led team Ctrl Alt Defend to **2nd Place** — **RootRush 2026 CTF** (Jeopardy-style Capture The Flag), Xploit Cybersecurity Club, IIIT Bhopal (NIIMACK 2026) — forensics, security analysis, and AI-based challenges.
- Open-Source Contributor** — **GSSoC 2026** (GirlScript Summer of Code): merged **10+ pull requests** across multiple community projects — bug fixes, new API endpoints, and performance/security improvements; earned 11 badges including Power Contributor (rare).
- Solved **70+ Data Structures & Algorithms problems** on LeetCode (Java) — arrays, strings, trees, and dynamic programming.